Mandatory Assignment 1 Testing

Note! JavaScript and Playwright were used to solve the tasks where test coding was required.

Task 1 Part 1 (Test automation with JavaScript and Playwright)

Task 1.1)

Navigate and be familiar with the following website:
https://parabank.parasoft.com/parabank/index.htm

Task 1.2)

In this task, navigate to the customer care page to fill out a form to check if things work correctly.

Task 1.3)

Test case (Customer care webpage):

Description: a user should get access to and send a message to the customer care service

Precondition: None

Assumption to this case: the user should easily access the webpage successfully from any browser, whether on MAC or Windows!

Test steps:
1. Navigate to https://parabank.parasoft.com/parabank/index.htm
2. Further, go to the "Customer care" page from the home page by clicking on the 'mail' icon. You'll see a form you need to fill out!
3. In the 'Name' field, enter a name. In the 'Email' field, enter an e-mail address. In the 'Phone' field, enter a phone number. In the 'Message' field, enter a message. (It's recommended to type in fake credentials, so be careful! We're not responsible for hacking attempts against you or any individuals out there 😊)

4. Click the 'Send to customer care' button.
The result: A page with the text 'Thank you 'name'. A Customer Care Representative will be contacting you.' Will appear on the screen. t

Task 1.4) Installed Playwright extension in VS Code.

Task 1.5) (Automated code from the Customer care test case in 1.3)

```
import { test, expect } from '@playwright/test';

test('Customer Care Test 2025', async ({ page }) => {
  //Go to the url page.
  await page.goto('https://parabank.parasoft.com/parabank/index.htm');
  await page.getByRole('link', { name: 'contact', exact: true }).click();
  // Fill out the form.

  await page.locator('#name').click();
  await page.locator('#name').fill('Test');
  await page.locator('#email').click();
  await page.locator('#email').fill('test@hahatest.com');
  await page.locator('#phone').click();
  await page.locator('#phone').fill('987654321');
  await page.locator('#message').click();

  // Write and submit the message.
  await page.locator('#message').fill('This is a test haha 2025');
  await page.getByRole('button', { name: 'Send to Customer Care' }).click();
});
```

Note! *To run the code , we typed in the following command in the VS Code terminal: "npx playwright test" , which automatically tests the code on the website.*

Task 1 Part 2

2.1

Other benefits we get from ensuring that our tests are independent of each other are:

1. Improved debugging, which makes it easier to locate the defects.
2. Automated test ensures the production of reliable results, which avoids the occurrence of false positives.
3. It has easier maintenance: independent tests are more straightforward to modify and maintain during the testing process.

2.2

Test Case 1:

Case Title: Bill payment on the service.

Description: New users should be able to register a new user from the main page: https://parabank.parasoft.com/parabank/index.htm

Test precondition: None

Assumption: The user is using a computer with a supported browser when opening the webpage.

Test steps:

1. Go to https://parabank.parasoft.com/parabank/index.htm
2. From the main page, click on the 'login'.
3. Click on Bill Pay link
4. Fill out the form (Again, use fake credentials).
5. Click on the "Send Payment" button.

Expected outcome: After successfully registering a user, a pop-up page should show up with a text seying the bill is successfully payed

Test case 2

Case Title: Transfer funds

Description: New users should be able to access the "Transfer Funds" page from the main page: https://parabank.parasoft.com/parabank/index.htm

Test precondition: The user should be logged in and registered on their services

Assumption: The user is using a computer with å supported browser when opening the webpage

Test steps:

1. Go to https://parabank.parasoft.com/parabank/index.htm
2. From the main page, click on the 'Transfer Funds' link.
3. Enter the desired amount, ex. $100 and the account money should be transferred.
4. Click on the "Transfer" button.

Expected outcome: After successfully registering a user, a pop-up page should show up with the text « Transfer Complete " as the title with more details about the transaction.

Test Case 3

Case Title: Para bank loan Request

Description: New users should be able to access the "Request loan" page from the main page: https://parabank.parasoft.com/parabank/index.htm

Test precondition: The user is logged in and registered to the Parabank service

Assumption: The user uses a computer with a supported browser when opening the webpage.

Test steps:

1. Go to https://parabank.parasoft.com/parabank/index.htm
2. After logging in, click on the 'Request Loan' button.
3. Enter the desired amount and downpayment amount , choose the account.
4. Click on the "Apply now" button.

Expected outcome: After successfully registering a user, a pop-up page should show up with the text " **Loan Request Processed".**

2.5 Code for the 1st test case (Bill Payment):

```
import { test, expect } from '@playwright/test';
test('test', async ({ page }) => {
  await
page.goto('https://parabank.parasoft.com/parabank/index.htm?ConnType=JDBC');
  await page.locator('input[name="username"]').click();
  await page.locator('input[name="username"]').fill('user');
  await page.locator('input[name="password"]').click();
  await page.locator('input[name="password"]').fill('test');
  await page.getByRole('button', { name: 'Log In' }).click();
  await page.getByRole('link', { name: 'Bill Pay' }).click();
  await page.locator('input[name="payee\\.name"]').click();
  await page.locator('input[name="payee\\.name"]').fill('tset');
  await page.locator('input[name="payee\\.name"]').press('Tab');
  await page.locator('input[name="payee\\.address\\.street"]').fill('haha');
  await page.locator('input[name="payee\\.address\\.street"]').press('Tab');
  await page.locator('input[name="payee\\.address\\.city"]').fill('lovers
streer');
  await page.locator('input[name="payee\\.address\\.city"]').press('Tab');
  await page.locator('input[name="payee\\.address\\.city"]').click();
  await page.locator('input[name="payee\\.address\\.city"]').fill('gang city');
  await page.locator('input[name="payee\\.address\\.city"]').press('Tab');
  await page.locator('input[name="payee\\.address\\.state"]').fill('state 1');
  await page.locator('input[name="payee\\.address\\.state"]').press('Tab');
```

```
  await page.locator('input[name="payee\\.address\\.zipCode"]').fill('87683');
  await page.locator('input[name="payee\\.address\\.zipCode"]').press('Tab');
  await page.locator('input[name="payee\\.phoneNumber"]').fill('987654321');
  await page.locator('input[name="payee\\.phoneNumber"]').press('Tab');
  await page.locator('input[name="payee\\.accountNumber"]').fill('1234567');
  await page.locator('input[name="payee\\.accountNumber"]').press('Tab');
  await page.locator('input[name="verifyAccount"]').fill('1234567');
  await page.getByRole('table').click();
  await page.locator('input[name="amount"]').click();
  await page.locator('input[name="amount"]').fill('12378');
  await page.getByRole('table').click();
  await page.getByRole('button', { name: 'Send Payment' }).click();
  await page.getByRole('link', { name: 'Log Out' }).click();
});
```

2.6

Code for the 2nd test case (Transferring funds)

```
import { test, expect } from '@playwright/test';

test('Transfer funds', async ({ page }) => {
  await page.goto('https://parabank.parasoft.com/parabank/index.htm);
  await page.locator('input[name="username"]').fill('alice');
  await page.locator('input[name="password"]').click();
  await page.locator('input[name="password"]').fill('alicew');
  await page.getByRole('button', { name: 'Log In' }).click();
  await page.getByRole('link', { name: 'Transfer Funds' }).click();
  await page.locator('#amount').click();
  await page.locator('#amount').fill('100');
  await page.getByRole('button', { name: 'Transfer' }).click();
  await page.getByRole('link', { name: 'Log Out' }).click();
});
```

Task 2.7-2.9 (3rd case (requesting a bank loan))

```
import { test, expect } from '@playwright/test';
```

```
test('request loan parabank', async ({ page }) => {
  await page.goto('https://parabank.parasoft.com/parabank/index.htm');

  await page.locator('input[name="username"]').click();
  await page.locator('input[name="username"]').fill('alice');
  await page.locator('input[name="password"]').click();
  await page.locator('input[name="password"]').fill('alicew');
  await page.getByRole('button', { name: 'Log In' }).click();
  await page.getByRole('link', { name: 'Request Loan' }).click();
  await page.locator('#amount').click();
  await page.locator('#amount').fill('10000');
  await page.locator('#downPayment').fill('600');
  await page.getByRole('button', { name: 'Apply Now' }).click();
  await page.getByRole('link', { name: 'Log Out' }).click();
});
```